

Adaptive Planar Search:

A Terminal-Placement Acquisition for Derivative-Free Optimization

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Draft, September 2026

Abstract

Derivative-free optimizers spend most of their budget on one-dimensional line searches. We use the terminal-placement rule of the companion theory paper as a line-search inner loop: with its scale and correlation length estimated online, it flees, probes, or commits in one to three evaluations. On the HumpDay physics benchmark it is a competitive inner loop on rough and high-dimensional objectives at tight budgets, where its early exits buy more search directions than a classical line search. A projection argument then lifts the same rule off the line: after two observations under any radial Gaussian-process prior, the entire next-point decision lives in a plane, in any ambient dimension. We give the exact planar placement and an adaptive planar search that follows from it.

1 Line searches are where the budget goes

Derivative-free optimizers reduce a high-dimensional problem to a sequence of one-dimensional subproblems: pick a direction, place a few evaluations along it, return a point. Classical inner loops — golden section, Brent — assume smooth unimodal slices and spend eight to ten evaluations resolving each one. On the rough slices of real physics objectives that assumption fails, and on a tight global budget the cost of each line is what binds.

The companion theory paper solves a different one-dimensional problem exactly: place three evaluations on an exponentiated Ornstein–Uhlenbeck slice and be paid the value at the final point. Its policy flees a below-median start, probes between two good values, and stays above a ceiling, all in closed form. Read as an inner loop it costs one to three evaluations per line, not eight to ten. This paper asks what that buys in an optimizer, and how far off the line the same geometry reaches.

Two findings. As an inner loop the rule is competitive on rough and high-dimensional objectives when the budget is tight, for a reason that is about evaluation economy more than about the placement theorem. And the placement geometry is not confined to a line: after two observations under a radial prior the next-point decision collapses to a plane, exactly, in any ambient dimension, which turns the line search into an adaptive planar search.

2 The terminal-placement inner loop

We implement the three-evaluation rule as an inner loop, **grass3**: the exact final placement and opening step of the theory paper, with the mean, scale, and correlation length estimated

Objective	d	grass3	rank	best rival	seed wins
plinko_funnel	7	-57.0	1/5	golden6 -50.8	13/16
wind_farm	16	-94.1	1/5	golden6 -93.6	16/24
bowling	4	-105.0	1/5	golden2 -104.0	4/8
free_kick	4	-32.9	5/5	golden2 -105.2	3/24

Table 1: **grass3** as an inner loop, median objective over seeds at budget 120; lower is better. Comparators are golden section (two and six evaluations) and Brent in the shared outer loop, and whole-cube random search. Rank is among the five, averaging tied medians; seed wins count seeds on which **grass3** beats the best rival. Across all 22 objectives **grass3** has mean rank 2.95/5 and is first or second on eight (nine counting joint second places); **free_kick** is a representative loss.

online from the optimizer’s own history. It flees or stays after one or two evaluations when the rule dictates and places the third otherwise. It is a heuristic adaptation, not the exact policy: it interpolates the opening-correlation table, extrapolates it above the tabulated range, forces flight after three stalled lines, and clips placements to the unit cube. The last is a genuine approximation — choosing an unconstrained exterior winner and then clipping can lose to the best feasible interior point — and we flag it rather than hide it.

We benchmark **grass3** inside a common outer loop, iterated random-direction search at budget 120, against four fixed inner loops: golden section with two and with six evaluations, Brent, and random placement, on matched directions and seeds.

3 Benchmark

The objectives are the application suite of the HumpDay package (Cotton 2021): small physics and engineering problems in which a point in the unit cube sets a deterministic simulator’s controls and the objective is the outcome — a break in pool, pins felled in bowling, a trebuchet’s range, a wind-farm layout. We use the 22 dynamics-simulating problems; their one-dimensional slices run from smooth to collision-cascade rough.

The rule wins on two kinds of objective. On collision and agent-chaos objectives with rough but structured slices it is the best inner loop; on a plinko objective it leads its strongest rival on 13 of 16 seeds (Table 1). It also wins on high-dimensional objectives even when slices are smooth: on a sixteen-dimensional wind-farm layout it leads golden6 on 16 of 24 seeds, and golden2 and Brent on 21 and 22.

The high-dimensional win is evaluation economy. Early exits of one or two evaluations let the outer loop try several times more directions than an inner loop spending eight to ten per line. The win is budget-starved rather than general: it appears when the dimension times a classical line’s cost exceeds the whole budget, so a method with a fixed number of directions per iteration cannot complete one pass.

The rule loses in two regimes. On landscapes too rough to carry navigable structure, broad sampling wins and placement theory has nothing to exploit; on needle landscapes, where the value hides in a spike, golden section’s fixed full-segment probes are the right move. It is also not a drop-in for the line search inside an existing optimizer: substituted into Powell, whose direction updates need genuine line minimization, it loses to Brent.

Three qualifications keep the reading honest. Random search samples the whole cube rather than sharing the outer loop, and the Brent baseline spends one call reevaluating the known incumbent, so the economy margin is somewhat overstated. An ablation is more sobering: omitting the third placement altogether helps on twelve of the twenty-two objectives, sometimes by a wide margin, so the benchmark speaks for evaluation economy and online adaptation more than for the placement theorem in particular. A control on exact exponentiated-OU fields, the model’s own landscape, loses to broad probing — terminal-placement wins do not come from the objectives resembling an OU path.

As a standalone optimizer rather than an inner loop, a budget-truncated two-shot form is mid-field against a panel of eleven standard methods, best at the smallest budgets and overtaken by population methods as the budget grows — the expected place for a rule built from a few-evaluation terminal-placement model.

4 From lines to planes

The line is not special. Fix a Gaussian-process prior with constant mean and radial covariance $K(\|x - y\|)$, and observe values at $x_1, \dots, x_n$. Let H be their affine hull and write a candidate as $x = h + w$ with $h \in H$ and $w \perp H$. Then

$$\|x - x_i\|^2 = \|h - x_i\|^2 + \|w\|^2, \quad (1)$$

so the candidate’s posterior depends only on its position within H and its distance from H . Every perpendicular direction is equivalent.

Proposition 1 (Dimension reduction). Under a radial Gaussian-process prior on an unrestricted domain, a single-point acquisition can be optimized in at most n dimensions after n observations, regardless of the ambient dimension.

With two observations the next-point decision lives in a plane, in any ambient dimension, and needs no OU structure. The companion theory paper computes that planar decision exactly for the exponential kernel: after observing values b, d at correlation ρ , the candidate’s correlations (x, y) to the two observations range over the triangle inequalities $\{xy \leq \rho, x \geq \rho y, y \geq \rho x\}$, whose boundary is the one-dimensional bracket and two exterior rays and whose interior holds one extra candidate, the point that matches both observed values.

That extra candidate is worth a measurable amount. With $b = d = \rho = \frac{1}{2}$ the optimal third point forms an equilateral triangle with the two observations, and its conditional expected terminal reward exceeds the best collinear choice by about 2.9%. The gain is a planar effect: it is unavailable to any line search.

5 An adaptive planar search

The reduction suggests a move. Evaluate a trial along a chosen direction; use its observed value to choose the next point within a plane containing that line; under the isotropic model, take any perpendicular direction. The plane costs one extra coordinate and, by the dimension-reduction proposition, captures the whole next-point decision no matter how large the ambient space.

dimension	random plane	momentum plane
$d = 2$	0.55	0.47
$d = 5$	0.28	0.23
$d = 10$	0.28	0.19

Table 2: Fraction of seeds on which the off-line (planar) third placement beats the on-line one, at budget 120 over 24 seeds and ten objectives. *Random plane*: the transverse direction is a random perpendicular. *Momentum plane*: it is a running estimate of the improving direction, projected off the line. In two dimensions the transverse direction spans the rest of the space and the off-line move mildly helps; in higher dimensions it is wasted, and a cheap direction estimate makes it worse, not better.

One adjustment is necessary, and the opening constant closes. The theory paper rewards the terminal recommendation, whereas an optimizer keeps the best evaluated point, so its opening maximizes $\mathbb{E}[\max(e^b, e^X)]$ over the probe correlation ρ , with $X \sim N(b\rho, 1 - \rho^2)$. The geometric reduction is unaffected — it is a fact about the posterior, not the objective — but the opening coefficient is not.

Proposition 2 (Expected-improvement opening). For the best-evaluated objective the optimal opening correlation satisfies $\rho^*(b) = \kappa_{\text{EI}} b + O(b^3)$ as $b \downarrow 0$, with

$$\kappa_{\text{EI}} = \frac{\mathbb{E}[e^Z; Z > 0]}{\mathbb{E}[Ze^Z; Z > 0]} = \frac{\Phi(1)}{\Phi(1) + \phi(1)} = 0.776639\dots, \quad Z \sim N(0, 1). \quad (2)$$

The first-order condition $\partial_\rho \mathbb{E}[\max(e^b, e^X)] = \mathbb{E}\left[e^X \mathbf{1}\{X > b\} (b - \rho X' / \sqrt{1 - \rho^2})\right] = 0$, with $X = b\rho + \sqrt{1 - \rho^2}X'$ and $X' \sim N(0, 1)$, gives at $b \downarrow 0$ the ratio $\mathbb{E}[e^Z; Z > 0] = \kappa_{\text{EI}} \mathbb{E}[Ze^Z; Z > 0]$, and $\mathbb{E}[e^Z; Z > 0] = e^{1/2}\Phi(1)$, $\mathbb{E}[Ze^Z; Z > 0] = e^{1/2}(\phi(1) + \Phi(1))$. So the optimizer opens *wider* than the terminal rule — $\kappa_{\text{EI}} = 0.777$ against the theory paper’s $\kappa_0 = 0.540$ — because a fresh high draw is worth more when the best point is kept rather than a mean recommendation. And it is elementary, where κ_0 is only characterized by a transcendental equation. A full planar-search implementation is the remaining empirical step.

6 Does the off-line placement help?

A budget-matched experiment isolates the planar move. From the incumbent we take two evaluations per step along a shared random direction; the only difference between strategies is where the second goes. *Line* keeps it on the line — extrapolate past the trial when it improved, reflect when it did not. *Planar* puts it at the equilateral apex over the base in a random plane containing the line. Start point, direction stream, and adaptive step are identical, so the comparison is on-line against off-line placement at the same budget, on ten HumpDay classic objectives over 24 seeds.

The reduction is exact for the posterior, but exploiting it is not free (Table 2). With a random plane the off-line placement wins on 55% of seeds in two dimensions and 28% by five, no better than before. The gap is the isotropy assumption cashing out: any perpendicular is equivalent *under the prior*, but a random perpendicular is almost surely not where the objective’s structure lies, so in high dimension the off-line evaluation is spent on an uninformative

direction. The exception proves the reading — on `drop_wave`, a radially symmetric ripple, the off-line move wins on 16–17 of 24 seeds in every dimension, because there the transverse direction always carries signal.

A cheap fix does not close the gap. Pointing the plane along a running momentum estimate of the improving direction is *worse* than a random perpendicular at every dimension (0.47, 0.23, 0.19): from a handful of points the estimate is poor, and committing the one off-line probe to a bad direction loses to sampling broadly. The reduction is exact, but aiming the plane is itself the hard problem — it wants a fitted subspace model, as in LineBO, not a momentum heuristic. That is the honest boundary of the result: the geometry hands high-dimensional search a plane for free, and the value of the plane is only as good as the direction one can afford to estimate.

7 Related work

LineBO (Kirschner et al. 2019) optimizes through adaptively selected one-dimensional subspaces; the reduction here identifies when one extra transverse coordinate carries measurable value, sharpening the choice of subspace to a plane after two observations. The Stochastic Three Points method (Bergou et al. 2020) compares an incumbent with two opposite trials; the planar candidate suggests an adaptive triangle variant whose third location is chosen after seeing the second value. Knowledge gradient (Frazier, Powell and Dayanik 2009) supplies the measure-then-choose framework; the contribution here is explicit geometry and the resulting planar move within it.

8 Scope

The defensible claim for high-dimensional derivative-free optimization is an exact simplification of the next decision under a radial prior: whatever the ambient dimension, two observations reduce it to a plane. Learning useful directions over a long run remains hard, and constraints or unknown anisotropy break the isotropy the reduction uses. The immediate application is therefore small evaluation budgets and local searches, where a single well-placed plane is most of what a step can afford.

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